

Consider a water reservoir system wherein river inflow and direct rainfall are the two (2) inputs, while the output would be the water discharge for different purposes, as shown in Figure 1. Then, answer the following items.

a. Give one (1) possible simulation that can be performed in this type of system. (5 points)

In this kind of scenario wherein the river inflow and direct rainfall are two inputs, while the output be the water discharge for different purposes, one possible simulation would be an Physical Simulation. This type of simulation creates a real or scaled-down physical model of the actual water reservoir system. It involves experimenting with this prototype by applying real inputs, such as river inflow and rainfall, and then observing the outputs like water discharge and losses. Physical simulation helps in understanding how the reservoir behaves in different conditions by directly measuring physical changes. It is useful because it closely mimics the real system and can validate other types of simulations, giving confidence in the design and operation of the reservoir system.

b. Identify at least three (3) components that you should consider in your specified simulation and justify why it is important. (5 points each)

- Physical Prototype of the Reservoir
This is the actual scaled model of the reservoir where you conduct experiments to understand flow and storage.
- Input Controls for River Inflow and Rainfall
These allow you to simulate real environmental inputs, important for testing how varying water sources affect reservoir performance.
- Measurement Tools for Seepage and Evaporation Loss
Instruments that measure water loss in the prototype are crucial to accurately assess and analyze the net water available.

C. What characteristics of simulation can possibly be achieved in your stated simulation? Rationalize your answer. (5 points)

- Produces realistic and reliable results by working with an actual model
- Allows direct observation and measurements of processes and outcomes
- Useful for testing physical changes or design modifications in a controlled environment
- Helps validate assumptions made in computer or numerical models
- Can demonstrate the effects of inputs and losses visually and tangibly

